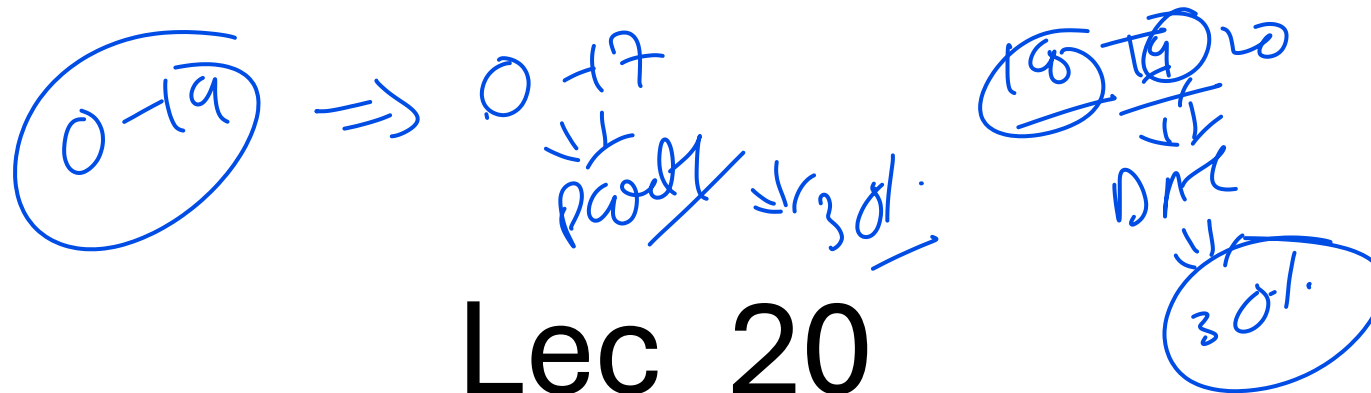




Lec_20

Solving Recurrence relation

1. Substitution method

18 $\Rightarrow$ DMC, recursion for given

19 $\Rightarrow$ RR DM

$$T(n) = T(n-1) + n^2$$

find T_c and SF

(ST)

✓ Prove by induction

(ST)

$$T(n) = \begin{cases} 1, & \text{if } n=1 \\ T(n-1)+n, & \text{if } n \geq 1 \end{cases}$$

1.) $T(n)=1$, if $n=1$

else $T(n-1)+n, n \geq 1$

$$T(n) = T(n-1) + n + c, \quad n \gg c$$

Big

Big $\gg$ small
 $\downarrow$
 ignore

$$T(n) = T(n-1) + n$$

2n

$$T(n) = T(n-1) + n$$

$$T(n) = T(n-2) + (n-1) + n$$

$$T(n) = T(n-3) + (n-2) + (n-1) + n$$

$$T(n) = T(n-4) + (n-3) + (n-2) + (n-1) + n$$

$$= 1 + 2 + 3 + 4 + \dots$$

$$(n-3) + (n-2) + (n-1) + n$$

n=5

$$\begin{aligned} T(n-1) &= T(n-2) + (n-1) \\ T(n-2) &= T(n-3) + (n-2) \\ T(n-3) &= T(n-4) + (n-3) \end{aligned}$$

$n=1$

$$T(n) = 1$$

$n-4 \leq 1$

$$T(n) = n + (n-1) + (n-2) + (n-3) + \dots + 3 + 2 + 1$$

$$T(1) = 1$$

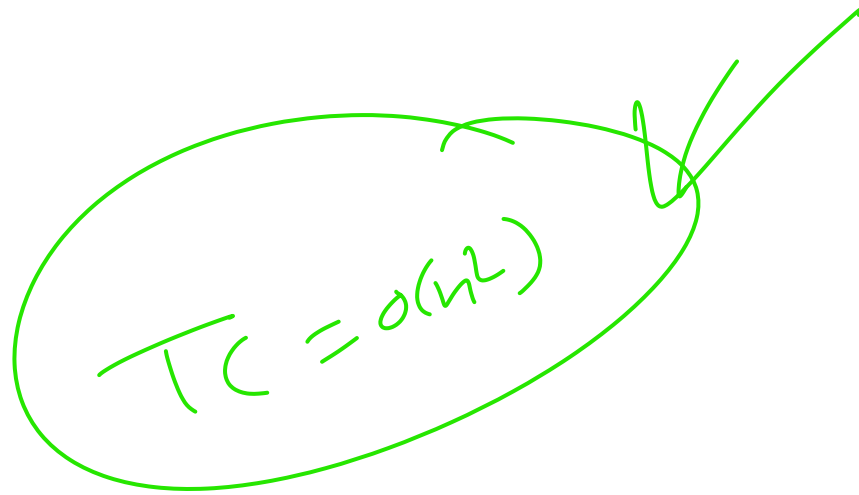
$T(n) =$ Sum of n natural Number from 1 to n

$$= \frac{n(n+1)}{2}$$

$$\frac{n^2 + n}{2}$$

Ans

$$T(n) = O(n^2)$$

A hand-drawn green oval containing the text $TC = O(n^2)$. A green checkmark is drawn over the right side of the oval.
$$TC = O(n^2)$$

2.) $T(n) = 1$, if $n=1$ small
 else $T(n-1) * n$, $n \geq 1$ Big

$$T(n) =$$

$$\text{if } n = 1$$

$$T(n) = C + n T(n-1)$$

$$T(n) = \begin{cases} 1, & \text{if } n = 1 \\ n T(n-1), & \text{if } n \geq 1 \end{cases}$$

$$1 \cdot n(n-1) \cdot n(n-2) \cdot n(n-3) \dots$$

$$2 n(n-1)$$

$$T(n) = n T(n-1) + C$$

$$\begin{aligned} T(n-1) &= (n-1) T(n-2) + C \\ T(n-2) &= (n-2) T(n-3) + C \\ T(n-3) &= (n-3) T(n-4) + C \end{aligned}$$

By Substitution

$$\begin{aligned} T(n) &= n T(n-1) + C \\ T(n) &= n (n-1) T(n-2) + C + C \end{aligned}$$

$$T(n) = n * (n-1) * (n-2) * T(n-3) + \underline{\underline{C + C + C}}$$

$$T(n) = \underbrace{n * (n-1)}_{\text{green}} \underbrace{(n-2)}_{\text{green}} \underbrace{(n-3)}_{\text{green}} \underbrace{T(n-4)}_{\text{green}} + \underbrace{C + C + C}_{\text{green}} \quad n-4=1$$

$$T(n) = \underline{n * (n-1) * (n-2) * (n-3) * (n-4) * (n-5)} = \dots * 3 * 2 * 1$$

$$\cancel{TC = O(n!)}$$

$$n! + C$$

$$TC =$$

$$n! + C$$

$$\underline{O(n!)}$$

3.) $T(n) = 1$, if $n=1$

else $T(n-1) + \log(n)$, $n \geq 1$

small

$$T(n) = T(n-1) + \log n + C$$

big

$$T(n) = T(n-1) + \log n$$

$$\begin{aligned} T(n-1) &= T(n-2) + \log(n-1) \\ T(n-2) &= T(n-3) + \log(n-2) \\ T(n-3) &= T(n-4) + \log(n-3) \end{aligned}$$

$n=1$

$$T(n) = T(n-1) + \log n$$

$$= T(n-2) + \log(n-1) + \log n$$

$$= T(n-3) + \log(n-2) + \log(n-1) + \log n$$

$$= T(n-4) + \log(n-3) + \log(n-2) + \log(n-1) + \log n$$

$$= \log n + \log(n-1) + \log(n-2) + \dots + \log(1)$$

$$T(n) = \log(n) + \log(n-1) + \log(n-2) + \dots + \log(1)$$

$$\log(abcd) = \log a + \log b + \log c + \log d$$

$$\log 2 + \log 2$$

$$T(n) = \log(n \times (n-1) \times (n-2) \times (n-3) \times \dots \times 1)$$
$$= \log(n!)$$

$$T(n) = \log n!$$

$$T(n) = O(\log n!)$$

Thanks

